Advanced JavaScript

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Day2

These are the Golden Days of JavaScript

JavaScript is
Multi-paradigm
Programming Language.

JavaScript supports programming in many different styles.

Object-Oriented JavaScript

Object-Oriented JavaScript

- The main principle with OOP is the use of Classes to create objects, and that objects are implemented in a manner that allows them to adopt Inheritance, Polymorphism, and Encapsulation.

- In most other object-oriented languages you would instantiate an instance of a particular class, but that is not the case in JavaScript.

- Unlike most other object-oriented languages, JavaScript doesn’t actually have a concept of classes. It looks and behaves differently.

Object-Oriented JavaScript

- JavaScript is a *class-free*, object-oriented language
- Although ES6 introduces JavaScript class expressions and class declarations, to provide a much clearer syntax to create objects and deal with.
- In fact classes are functions
- Custom Object that you, as a JavaScript developer, create and use is the main actor in application.

Custom Object

- Objects that you, as a JavaScript developer, create and use.

- An object in JavaScript is a complex construct usually consisting of a constructor as well as zero or more methods and/or properties.

- Objects can be either stand-alone with their own set of properties & functions or they can inherit properties from other objects

Custom Object

- There are different ways to create an instance of an object class (Functions in JavaScript)
  - Basic Object Literal Pattern
  - Factory Function
  - Custom Object Constructor Function
  - …

Literal Pattern Object Creation

```
var obj =  { };

obj.name = “banana”

obj.click = function(){
    alert( “you can eat" );
    }

obj.details ={
        mycolor: ”yellow”,
        mycount:12
        }

//(obj instanceof Object) // true
```

Literal Pattern Object Creation

```
var obj = {
// Set the property names and values use key/value pairs
    “name” : “banana”, // name : “banana”
    click : function(){
        alert( “you can eat" );
        },
    //initialize entire object
    details : {
            mycolor: ”yellow”,
            mycount:12
        }
};
```

Custom Object creation using basic object literal pattern

- **We can create objects with a short syntax that defines an object inside curly braces. (basic object literal pattern)**

```
var emp1 = { name:“Aly", age: 23};

var emp2 = { name: “Hassan",   age: 32};
```

Custom Object creation using basic object literal pattern

- After an object exists, you can add a new property to that instance by simply assigning a value to the property name of your choice.

- For example, to add a property about the “Salary” for “Hassan”, the statement is:

```
var emp1 = { name:“Aly", age: 23);
var emp2 = { name: “Hassan",  age: 32);
emp2.salary = 320;
```

- After that assignment, only emp2 has that property.

- There is no requirement that a property be pre-declared in its constructor or shortcut creation code.

Factory Function Pattern

- It is a way where object is created as a return of a function call assigned to a variable
- Used to create Multiple Objects with same interface
- No need to use “new” when calling a factory function

Creating New Instance from Custom Object using Factory Pattern

- Factory Function for Employee Object

```
var Employee = function (e_nm, e_ag){
    return {
        name : e_nm,
        age : e_ag
        }
}
```

```
var Employee = function (e_nm, e_ag) {
    var emp = { name : e_nm,
                age : e_ag
              };
    return emp;
}
```

- Creating object instances using Factory Function Method

```
var emp1 = Employee (“Aly", 23);
var emp2 = Employee (“Hassan", 32);
var emp3 = Employee ();
```

Constructor Function

- A constructor function looks like any other JavaScript function, but its purpose is:
  - to define the initial structure of an object
  - to define it's property and method names
  - It can populate some or all of the properties with initial values.
  - Values to be assigned to properties of the object are typically passed as parameters to the function,
  - Statements in the constructor function assign those values to properties.
- MyConstructor
- myFunction

Creating New Instance from Custom Object using Constructor Method

- Constructor Function for Employee Object

```
function Employee (name, age){
    this.name = name;
    this.age = age;
}
```

- To create object instances using Constructor Function Method, invoke the function with the new keyword

```
var emp1 = new Employee (“Aly", 23);

var emp2 = new Employee (“Hassan", 32);

var emp3 = new Employee ();
```

Adding methods to Constructor Function (Functional shared Pattern)

- Functional shared pattern is used to save memory by adding methods to the constructor function:

```
function Employee(name, age){
    this.name = name;
    this.age = age;
    this.show = showAll;
}

function showAll( ){
    alert("Employee " + this.name + " is " + this.age + " years old.");
}
```

Property

Method

Adding methods to Constructor Function (Functional Class Pattern)

- Adding methods to the constructor function using Function Literal:

```
function Employee(name, age) {
    this.name = name;
    this.age = age;
    this.show =  function () {
        alert("Employee " + this.name + " is " + this.age + " years old.");
    }
}
```

Property

Function Literal

Instance Object Creation

```
// Class using constructor function
function User( name ) {
    this.name = name;
        this.display = function(){return this.name;}
}

// Instance object of user
var me = new User( "My Name" );

// Test
alert( me.name );
alert( me.display());
alert( me.constructor == User);//true
alert( me.constructor == Object);// false
```

Reminder:
Function Default arguments

```
function myFun(){
	var x = arguments[0] || 10;
	var y = arguments[1] == undefined ? 11 :
arguments[1]

	return x + y;
}



myFun(); //21
myFun(1); //12
myFun(1,2); //3
```

Creating New Instance from Custom Object via Constructor Overloading

- Assign a *default value* to a Property:

```
function Employee (id='idx' /*ES6*/),name, age,salary=2000/*ES6*/){
    this.name = typeof name ==="undefined"? "Nour": name;
    this.age = age|| 0; //ES5
    this.salary = salary;
    this.id = id;
}
```

- We can also generate a blank object and then populate it explicitly; property by property:

```
var emp1 = new Employee( );
emp1.name = "Aly";
emp1.age = 23;
```

Overloading

- A common feature in other object-oriented languages is the ability to “overload” functions.
- Overloading occurs when more than one method within the same class have the same method name but different in parameters (different numbers and/or types of passed arguments) to perform different behaviors
- Overloading can be fulfilled via
  - function arguments property using default parameters & any conditional statement.
  - Creating function that calls the meant function with proper requirement

Constructor Function, new & this

- When function invoked with new, functions return an object known as this.
- "new" before any function call turns it into constructor function
- JavaScript uses "this" keyword to refer to the current object.
- "this" is confusing sometimes, when it doesn't return the expected object.
- You have a chance of modifying this before it is returned

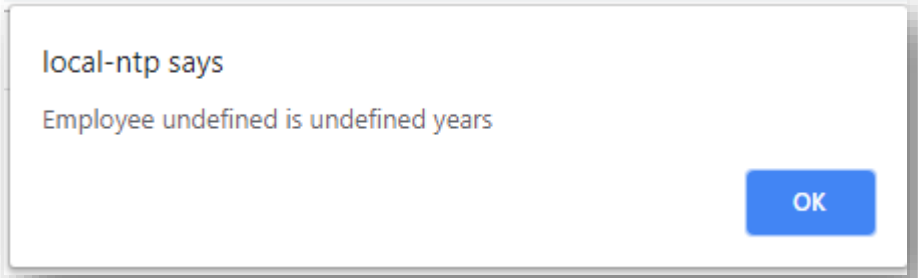
“new” Operator

- When using “new”
  - A brand new empty object is created
  - That object get linked to another object
  - It gets bound as “this” keyword as a purpose for function call
  - If the function doesn’t return any thing, it will return “this”.

- Note:
  - Primitive datatypes pass by value while Objects and Arrays pass by reference;
    - When any change happens in obj1 it is reflected in obj2
  - using new can over come this problem

Example: “this” and Closure

```
function Employee(nm, age){
    this.eName = nm;
    this.age = age;
    this.show =  function () {

        setTimeout(function (){
            alert("Employee " + this.eName + " is " + this.age + " years”);
        },5000);
    }
}
```




```
var me = new Employee(“Nour”,5);
me.show()
```

?

“this” keyword & Binding

- Every function while executing, has a reference to its execution context called “this”.
- “this” is an identifier that gets the value of object bound to it, it behaves like normal parameters.
- “this” binding is dependent on its “call site” (where the function get executed)

“this” keyword

- “this” is dynamic since it looks for things at runtime, based upon how you call things

- 4 rules for binding “this” (in terms of order precedence) depending on call site
  - Default Binding
  - Implicit Binding
  - Explicit Binding
  - new keyword

Hard Binding

Default & Implicit Binding

- Default Binding
  - It is applied on a standalone functions & IIFEs
    - Function defined in Global Scope
  - Depends on Strict Mode of code running inside a function
    - Its value is undefined in strict mode,
      - To be applied globally should be called as window.fn
    - otherwise its value is Global Object

- Implicit Binding
  - An object is calling the function
    - Object on the left of the (.) function call

Example

```
function myFun(){
    console.log(this.val)
}
var val = “myVal”;

var myObj1 = {val : ”obj1Val”, myFun: myFun};

var myObj2 = {val : ”obj2Val”, myFun : myFun};

myFun(); //myVal
myObj1.myFun(); //obj1Val
myObj2.myFun(); //obj2Val
```

Explicit Binding

- It’s a hard binding
- When function is called, it predict its object
- If you want to set a specific object other than the calling object make hard binding using Function Object methods.
  - bind()
  - apply()
  - call()

Example!

Using call() and apply()

```
var myObj={
      name:"myObj Object",
      myFunc:function(){
                  alert(this.name)
            },
      myFuncArgs:function(x,y){
                  alert(this.name+" " + x +" "+y)
            }
};

var obj1={name:"obj1 Object"};
```

```
myObj.myFuncArgs(1,2);      //myObj Object 1 2

myObj.myFuncArgs.apply(obj1,[1,2]);     //obj1 Object 1 2

myObj.myFuncArgs.call(obj1,1,2);      //obj1 Object 1 2
```

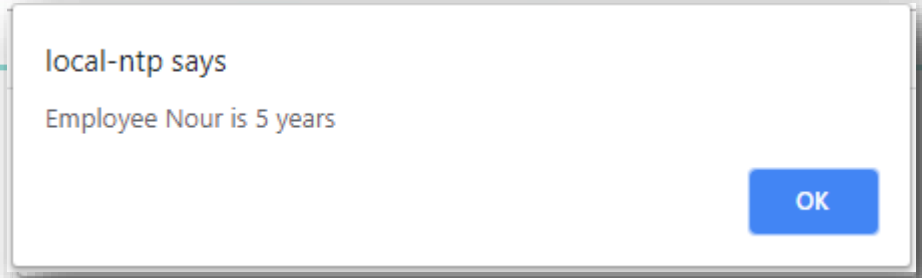
Using bind()

```
var myObj={
        name:"myObj Object”,
        myFunc:function(){
                        alert(this.name)
                },
        myFuncArgs:function(x,y){
                        alert(this.name+" " + x +" "+y)
            }
};

var obj1={name:"obj1 Object"};
```

```
myObj.myFuncArgs(1,2);  //myObj Object 1 2

myObj.myFuncArgs.bind(obj1)(5,6); //obj1 Object 5 6
myObj.myFuncArgs.bind(obj1,5)(6); //obj1 Object 5 6
myObj.myFuncArgs.bind(obj1,5,6)(); //obj1 Object 5 6
```

“this” and Closure

```
function Employee(name, age){
    this.eName = name;
    this.age = age;
    this.show =  function () {
        var that=this; //_that|_self|_this
        setTimeout(function (){
            alert("Employee " + that.eName + " is " + that.age + " years”);
        },5000);
    }
}
```




```
var me = new Employee(“Nour”,5);
me.show()
```

Reminder: “Scope” Basics

- A scope is the lifespan of a variable
- Programing languages have block and function scope
- In ES5; only functions have scope. Blocks (if, while, for, switch) do not have scope.
- ES6 presenting let for block scoping
- All variables are within the global scope unless they are defined within a function.
- All global variables actually become properties of the global object (window object).

Reminder: “Scope” Basics

- Variables inside a function are
  - Free var: if they are not declared inside function scope and belong to another scope
  - Bound var: if they are declared inside function
- In JavaScript function scope is lexical/static scope; where free variables belongs to parent scope
- Other language may have dynamic scope where free variables belongs to calling scope.
- JavaScript doesn't have dynamic scope
- When variables are *not declared using the var keyword,* they are declared globally.

Reminder: "Scope" Basics

```
var  myVar = "Hello"; // myVar is a global variable

// create function to modify its own myVar variable
function test (){
    var myVar = "Bye"; //Local variable to test()
     //this is called shadowing
}

test();


alert( myVar); // Global myVar still equals "Hello"
```

Shadowing
occurs when a scope declares a variable that has the same name as one in a surrounding scope; the outer variable is blocked in the inner scope

Privileged Method

- The term *privileged method* is not a formal construct, but rather a technique.
- It's coined by Douglas Crockford
- Privileged methods essentially have one foot in the door:
  - They can access private methods and values within the object
  - They are also publicly accessible

Privileged Method

```
var User = function (name, age) {
    var year = ((new Date().getFullYear() )- age);
    //year is a local variable → private member
    this.getYearBorn = function () { return year;};
};
// Create a new User
var user_1 = new User( “Aly”, 25 );
// Access privileged method to access private year value
alert( user_1.getYearBorn());
alert( user_1.year); // undefined because year is private
```

Private Methods

- Private methods are functions that are only accessible to methods inside the object and cannot be accessed by external code.

```
var User = function (name) {
    this.name = name;
        function welcome () {
        alert( "Welcome back, " + this.name + ".");}
    welcome();
}
// Create a new User
var me = new User( "Aly" ); // alerts: "Welcome back, Aly."
me.welcome();  // Fails because welcome is not a public method
```

Inner Function = Nested Function

Private Method

“this”, Closure and Private Method

```
function Employee(name, age,yr){
      this.eName = name;
      this.age = age;
      var yrbrn=yr;
     function welcoming() {
        alert("welcome " + this.eName + " you were born in " + yrbrn );
    }
    this.welExec=function(){

        return val();
    }
    //welcoming.call(this);
    var val=welcoming.bind(this) //val();
}
```

Hard binding

no matter what is the invocation context.

Make a function that calls internally and manually an explicit binding and force to do the same instruction

no matter where and how you invoke that function

Class Properties & Methods

- Class Properties and methods are similar to static properties and methods in other object oriented languages.

- This can be created by adding either property or method to a constructor function object.

- This is possible because functions in JavaScript are plain objects that can have properties and methods of their own.

Class Properties & Methods

```
function Employee(name, age){
        this.name = name;
        this.age = age;
}

Employee.count=0;

Employee.getCount=function(){
   return Employee.count
}
```

Example!

Assignments